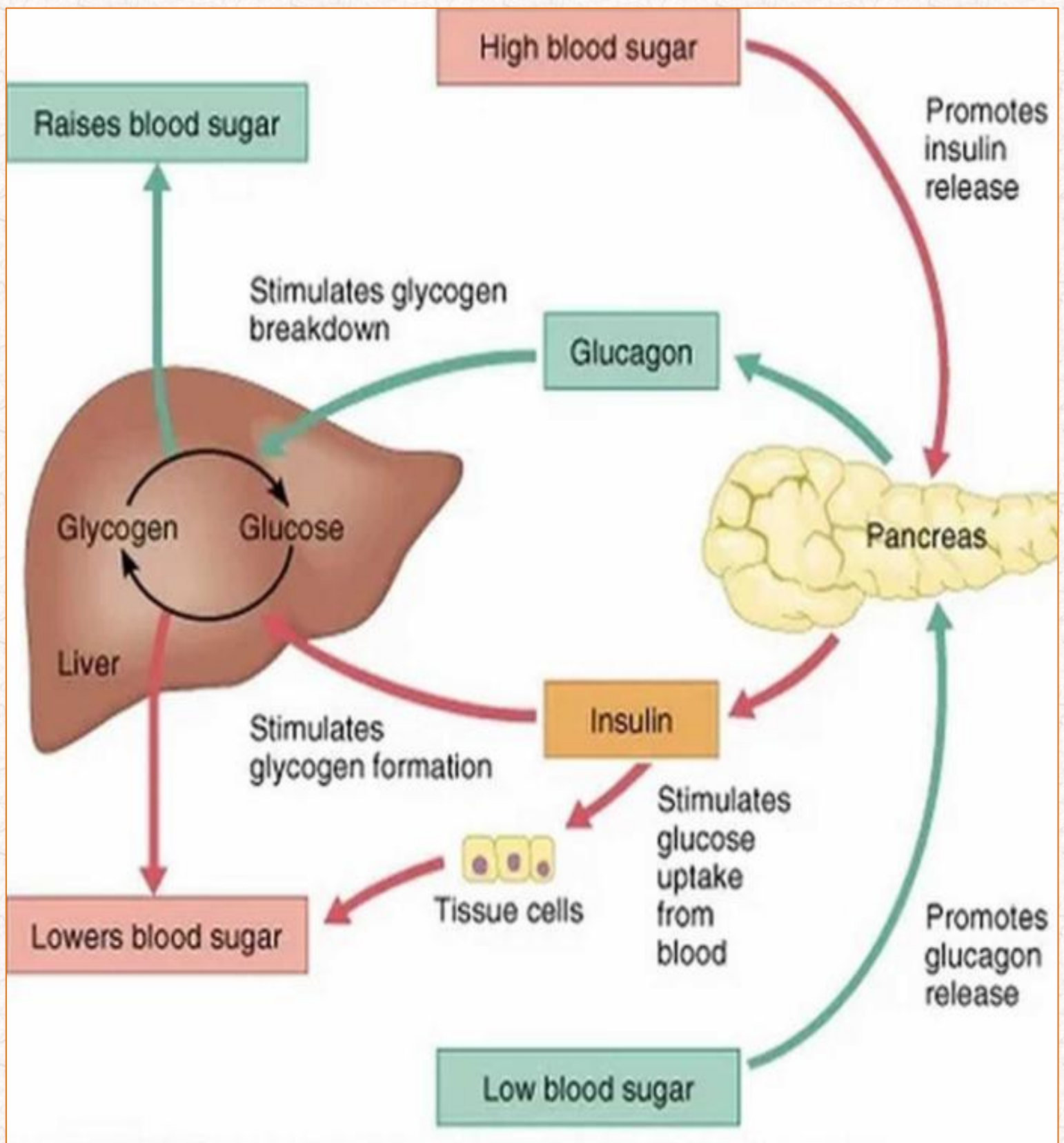


Outline

- ©Maintenance of blood glucose concentration
- ©Glycemic response to carbohydrates
- ©Integration metabolism in tissues

© Maintenance of blood glucose concentration

- Maintenance of normal blood glucose concentration is an important homeostatic function and is a major function of the **small intestine, liver, kidneys, skeletal muscle, and adipose tissue**.
- Regulation is the net effect of the organs' metabolic processes that remove glucose from or return glucose to the blood.
- These metabolic processes are hormonally influenced, primarily by the antagonistic pancreatic hormones **insulin** and **glucagon** and to a lesser extent by the **glucocorticoid** hormones of the adrenal cortex.
- The rise in blood glucose following the ingestion of carbohydrate, for example, triggers the release of insulin while reducing the secretion of glucagon.
- Insulin is the main hormone that lowers blood glucose levels and is the primary **anabolic** hormone.
- Insulin stimulates the cellular uptake of glucose, amino acids, and lipid, which leads to their conversion to storage forms in muscle and adipose tissue.
- The storage form for glucose, glycogen, is synthesized through the process called **glycogenesis**.
- Glucagon, the primary **catabolic** hormone that has opposite effects of insulin, increases the breakdown of liver glycogen by a process called glycogenolysis.
- Additional mechanisms to increase blood glucose levels include an increase in the secretion of glucocorticoid hormones, primarily cortisol.
- Glucocorticoids cause increased activity of hepatic **gluconeogenesis**,



Maintennance of blood glucose concentration

©Glycemic response to carbohydrates

- The effect that carbohydrate-containing foods have on blood glucose concentrations is called the **glycemic response to the food**.
- Some foods that are rapidly digested and absorbed (high-GI foods) cause a rapid rise in blood glucose levels that, because of the effect of insulin released in response, can subsequently lead to a rapid fall even below the fasting level.
- Other foods cause a slower and more extended rise with a lower peak level of glucose and insulin and a gradual fall (low-GI foods).
- The **glycemic index** is defined as the increase in blood glucose level above the baseline level (fasting level) during a 2-hour period following the consumption of a defined amount of carbohydrate (usually 50 g) compared with the same amount of carbohydrate in a reference food.
- The Glycemic Index of foods is ranked on a scale of 0-100.
- **Formula:**

There isn't a strict formula for calculating GI because it's determined experimentally, but the relative formula is:

$$GI = \frac{\text{Area Under the Blood Glucose Response Curve (AUC) for Test Food}}{\text{Area Under the Blood Glucose Response Curve (AUC) for Reference Food(glucose)}} * 100$$

Categories of GI:

©Low GI: 55 or less

©Medium GI: 56-69

©High GI: 70 or above

- The glycemic index concept was developed to provide a numerical value to represent the effect of a particular food on blood glucose levels.

- The **glycemic load** equals the glycemic index times the grams of carbohydrate in a typical portion of the food.
- Glycemic load, considers both the quantity and the quality of the carbohydrate in a food. Hence it gives a more accurate picture of a food's actual impact on blood sugar levels.
- Formula:
$$\frac{GI \times \text{Carbohydrate content in grams}}{100}$$

Categories of GL:

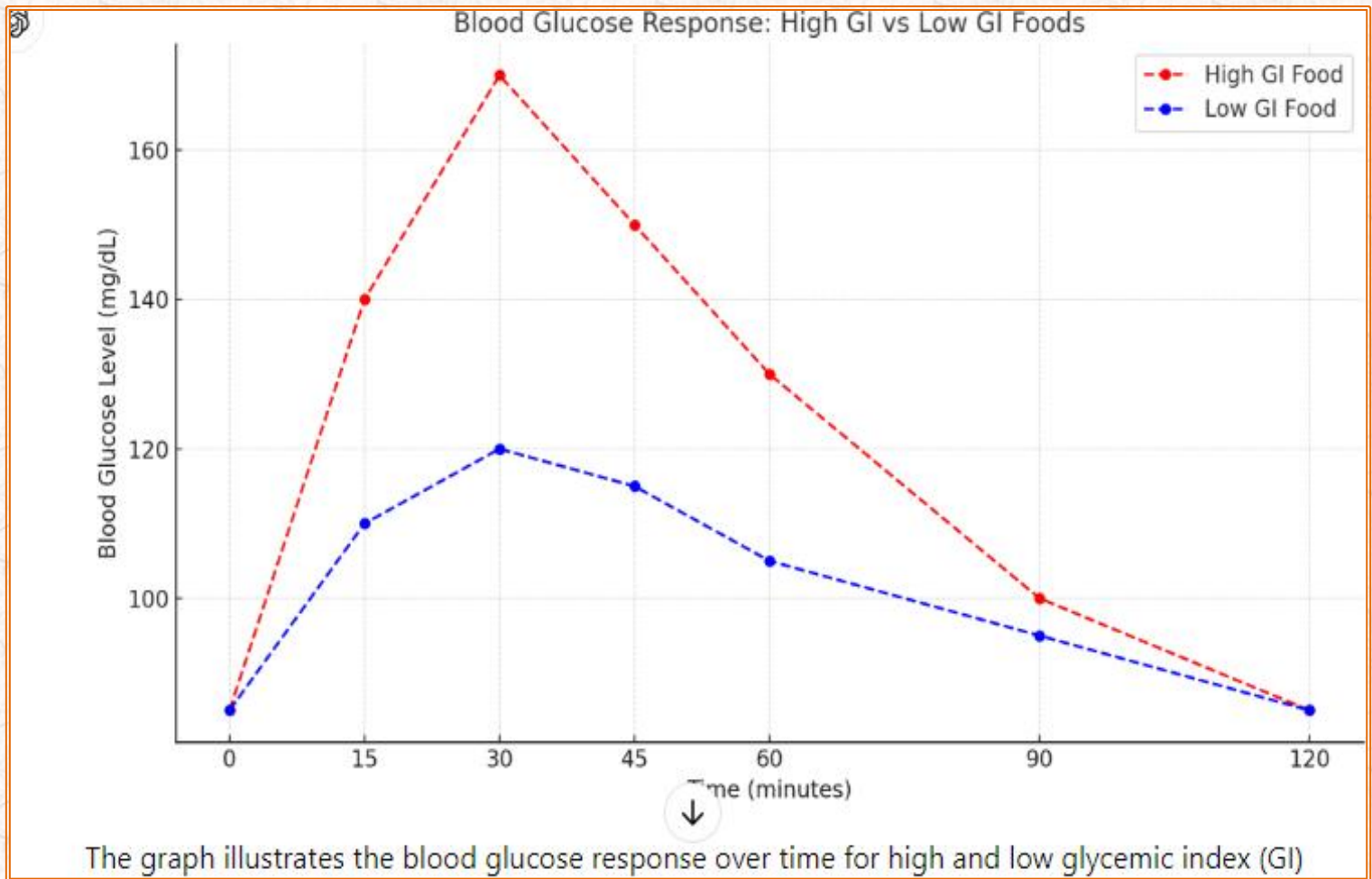
- ◎Low GL: 10 or less
- ◎Medium GL: 11-19
- ◎High GL: 20 or more

Differences Between GI and GL

Aspect	Glycemic Index (GI)	Glycemic Load (GL)
Focus	Quality of carbohydrates (speed of digestion)	Both quality and quantity of carbohydrates
Influence	Solely dependent on the type of carbohydrate	Dependent on GI and portion size
Usefulness	Useful for comparing foods with similar serving sizes	Better for estimating real-life blood sugar impact
Scale	Fixed scale (0–100)	Variable, depends on food quantity
Calculation	Experimentally measured	Calculated mathematically

Example

- A watermelon has a high GI (~72), but its GL for a typical serving is low (~4) due to its low carbohydrate content per serving. This demonstrates why GL provides a better understanding of a food's overall glycemic impact.



©High GI Food (Red Curve):

- Rapid increase in blood glucose levels.
- Peak at around 30 minutes, followed by a steep decline.
- This reflects fast carbohydrate digestion and absorption.

©Low GI Food (Blue Curve):

- Slower and more gradual rise in blood glucose levels.
- Lower peak and more sustained response over time.
- Indicates slower digestion and absorption of carbohydrates.

@Glycemic index and Glycemic load of some common foods

FOOD	Glycemic Load	Glycemic Index
	HIGH	
Raisins	28	64
Baked potato	26	85
White rice, boiled	23	64
Cornflakes	21	81
	Moderate	
Brown rice, boiled	18	55
White spaghetti, boiled	18	44
Wholemeal spaghetti, boiled	16	37
New potatoes, boiled	14	50
Porridge	13	58
Bulgur Wheat	12	48
Banana	12	52
Unsweetened orange juice	12	52
Parsnips	12	97
Sweet potato	11	44
Unsweetened apple juice	11	40
	LOW	
Honey	10	55
Sweetcorn, boiled	9	54
Wholemeal bread	9	71
Dried apricots	9	31
Muesli	9	56
Wholemeal rye bread	8	58
Grapes	8	46
Chickpeas	8	28
Mango	8	51
Fresh pineapple	7	59
Kidney beans, cooked	7	28
Kiwi fruit	6	53
Mixed grain bread	6	49
Apples	6	38
Oranges	5	42
Pears	4	38
Peas	3	48
Carrots	3	47
Whole Milk	3	27
Cashew nuts	3	22
Raw cherries	3	22
Peanuts	1	14

© For your further information, here is the scientific procedure how GI of a given test food is experimentally determined

- Calculating the Glycemic Index (GI) involves experimental measurement in a controlled environment. Here's a step-by-step procedure:

1. Selection of Participants

- **Number of Participants:** 10-12 healthy individuals without diabetes are typically recruited.
- **Health Status:** Participants should have stable blood sugar levels and no known metabolic disorders.

2. Preparation of Test Food

- **Amount of Carbohydrate:** Prepare a portion of the test food that provides 50 grams of digestible carbohydrates (excluding fiber).
- **Cooking and Serving:** Ensure the food is prepared and served in a consistent manner.

3. Preparation of Reference Food

- Use either glucose or white bread as the reference food.
- Provide an equivalent amount of carbohydrate (50 grams).

4. Fasting and Baseline Measurement

- Participants fast for 8-12 hours before the test.
 - Measure baseline blood glucose levels (time = 0) using capillary blood from a finger prick.
-

Food Consumption

- Participants consume the test food within 10-15 minutes.
 - Ensure the reference food is consumed in a similar manner on a separate day.
-

6. Blood Glucose Measurements

- Measure blood glucose levels at multiple time intervals: **15, 30, 45, 60, 90, and 120 minutes** after consumption.
 - Record the blood glucose levels for each time point.
-

7. Plot Blood Glucose Response Curve

- Plot the blood glucose levels (y-axis) against time (x-axis) for both the test food and the reference food.
 - The resulting curve represents the **Blood Glucose Response Curve (AUC)**.
-

8. Calculate Area Under the Curve (AUC)

- Determine the AUC for the test food and the reference food using the trapezoidal rule.
- Exclude the baseline (fasting) value from calculations.

9. Compute the GI

- Using the formula:
$$\frac{\text{AUC of the test food}}{\text{AUC of the reference food}} \times 100$$

10. Average the GI Values

- Calculate the average GI from all participants to determine the food's final GI value.

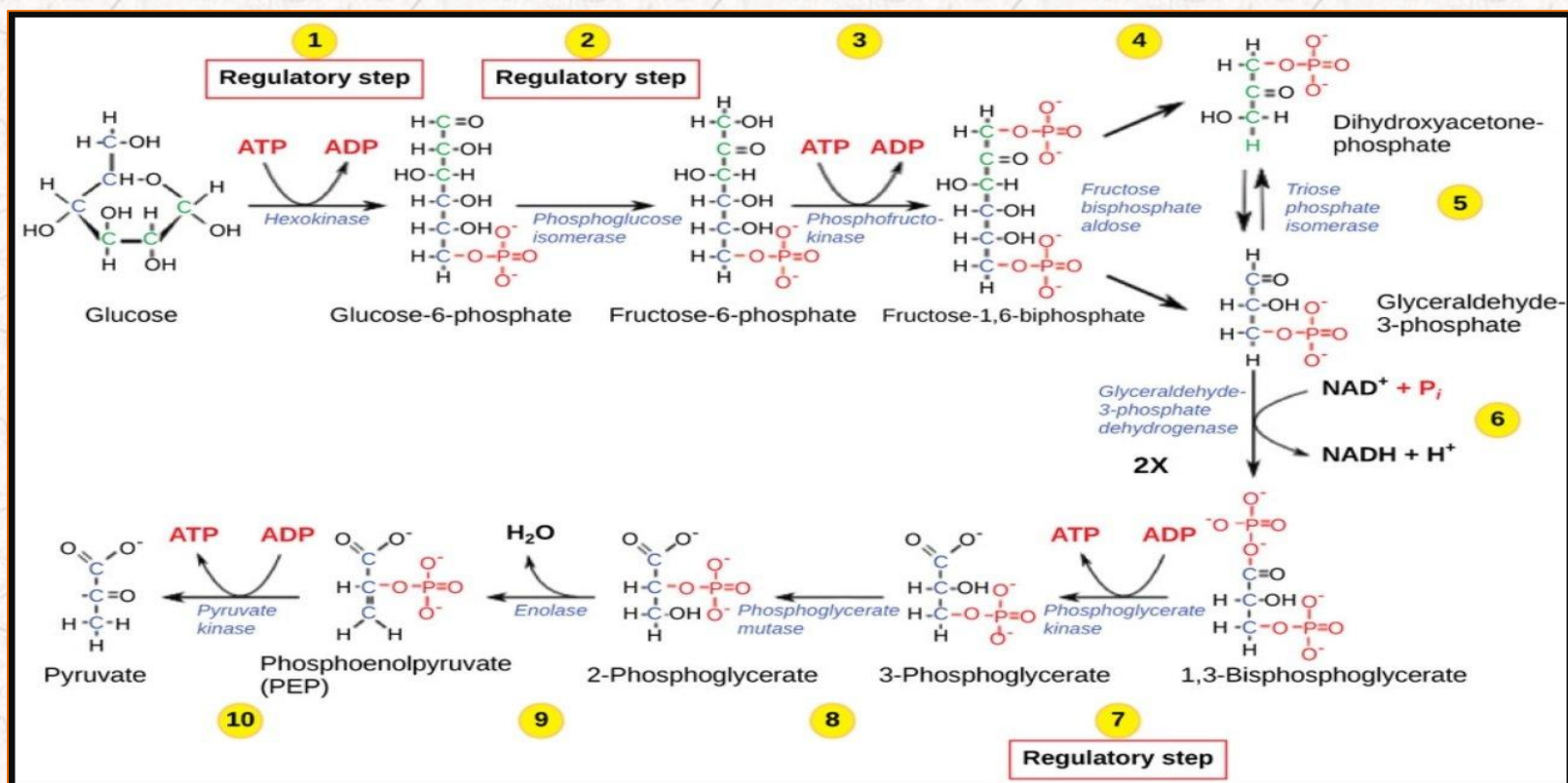
©INTEGRATED METABOLISM IN TISSUES

- The metabolic fate of the monosaccharides, especially glucose, depends to a great extent on the body's energy needs.
- The metabolic pathways of carbohydrate metabolism are :
- **Glycogenesis:** The synthesis of glycogen
- **Glycogenolysis:** The breakdown of glycogen
- **Glycolysis:** The oxidation of glucose to pyruvate
- **Gluconeogenesis:** The synthesis of glucose from noncarbohydrate sources
- **Pentose phosphate pathway (hexose monophosphate shunt):** The production of five-carbon monosaccharides (pentoses) and nicotinamide adenine dinucleotide phosphate (NADPH)
- **Tricarboxylic acid (TCA) cycle:** The oxidation of acetyl-CoA to yield CO_2 and high-energy electrons.

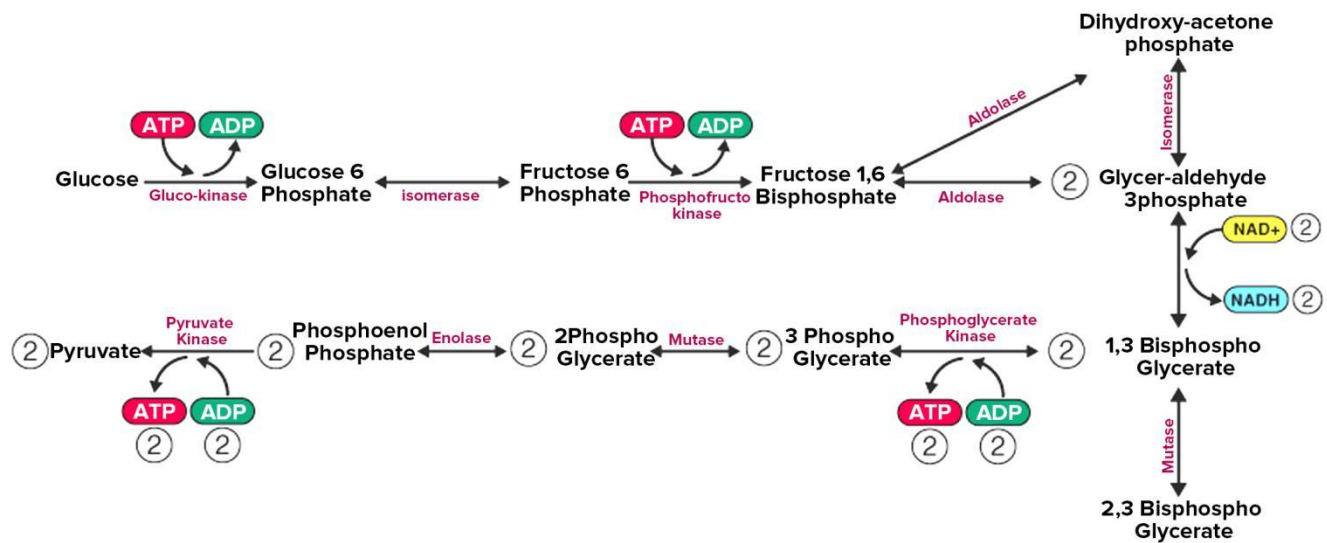
©Glycolysis

- Glycolysis is the pathway by which glucose is degraded into two 3-carbon units, pyruvate.
- From pyruvate, the metabolic course depends largely on the availability of reducing units in the cytosol, which is dependent upon the availability of oxygen within the cell.
- Glycolysis can function under either aerobic or anaerobic conditions. Under anaerobic conditions or in a situation without sufficient reducing equivalents due to either the lack of oxygen or high cellular metabolism, pyruvate is converted to lactate.
- Under otherwise normal conditions, the conversion to lactate occurs mainly in times of strenuous exercise when the demand for oxygen by the working muscles exceeds that which is available.
- Lactate produced under anaerobic conditions can also diffuse from the muscle to the bloodstream and be carried to the liver for conversion to glucose.
- Under these anaerobic conditions, glycolysis releases a small amount of usable energy that can help sustain the muscles even in a state of oxygen debt.

- Providing this energy is the major function of the anaerobic pathway of glucose to lactate.
- Anaerobic glycolysis is the sole source of energy for erythrocytes because these cells do not contain mitochondria.
- Both the brain and gastrointestinal tract also produce much of their energy from glycolysis.
- Under aerobic conditions, pyruvate can be transported into the mitochondria and participate in the TCA cycle, in which it becomes completely oxidized to CO_2 and H_2O .
- Complete oxidation is accompanied by the release of relatively large amounts of energy, much of which is captured in ATP molecules by the mechanism of oxidative phosphorylation.
- The glycolytic enzymes function within the cytosol of the cell, but the enzymes catalyzing the TCA cycle reactions are located within the mitochondrion.
- Therefore, pyruvate must enter the mitochondrion for complete oxidation.
- Glycolysis followed by TCA cycle activity (aerobic catabolism of glucose) demands an ample supply of oxygen, a condition that generally is met in normal, resting mammalian cells.
- In a normal, aerobic situation, complete oxidation of pyruvate generally occurs, with only a small amount of lactate being formed. The primary importance of glycolysis in energy metabolism, therefore, is in providing the initial sequence of reactions (to pyruvate) necessary for the complete oxidation of glucose by the TCA cycle, which supplies relatively large quantities of ATP.



Simplified path way of glycolysis



©Biological importance of glycolysis

supports Cellular Function in Hypoxic Conditions

- Cells in low oxygen environments, such as cancer cells and certain tissues under hypoxia, rely heavily on glycolysis for ATP production (**Warburg effect** in cancer).

Essential in Red Blood Cells

- Since red blood cells lack mitochondria, glycolysis is their sole source of ATP to maintain ion pumps and cellular integrity.

Link to Aerobic Respiration

- The end product of glycolysis, **pyruvate**, enters the mitochondria under aerobic conditions and is converted into acetyl-CoA, feeding into the **citric acid cycle** (TCA cycle) for further energy production.

Energy Production

- Glycolysis is the primary pathway for the catabolism of glucose, yielding energy in the form of **ATP**.
- It provides energy quickly, especially in cells with high energy demands, such as muscle cells during intense activity.

Universal Pathway

- Glycolysis occurs in almost all living organisms and does not require oxygen (**anaerobic process**), making it essential for energy production in anaerobic conditions or in organisms lacking mitochondria (e.g., red blood cells).

Precursor for Biosynthesis

- Glycolysis produces intermediates used as precursors for other biosynthetic pathways:
 - **Glucose-6-phosphate**: Used in the pentose phosphate pathway for nucleotide and NADPH synthesis.
 - **Fructose-6-phosphate**: Used for amino sugar production.
 - **Dihydroxyacetone phosphate (DHAP)**: Involved in triglyceride and lipid biosynthesis.
 - **3-phosphoglycerate**: Precursor for serine biosynthesis

Maintains Redox Balance

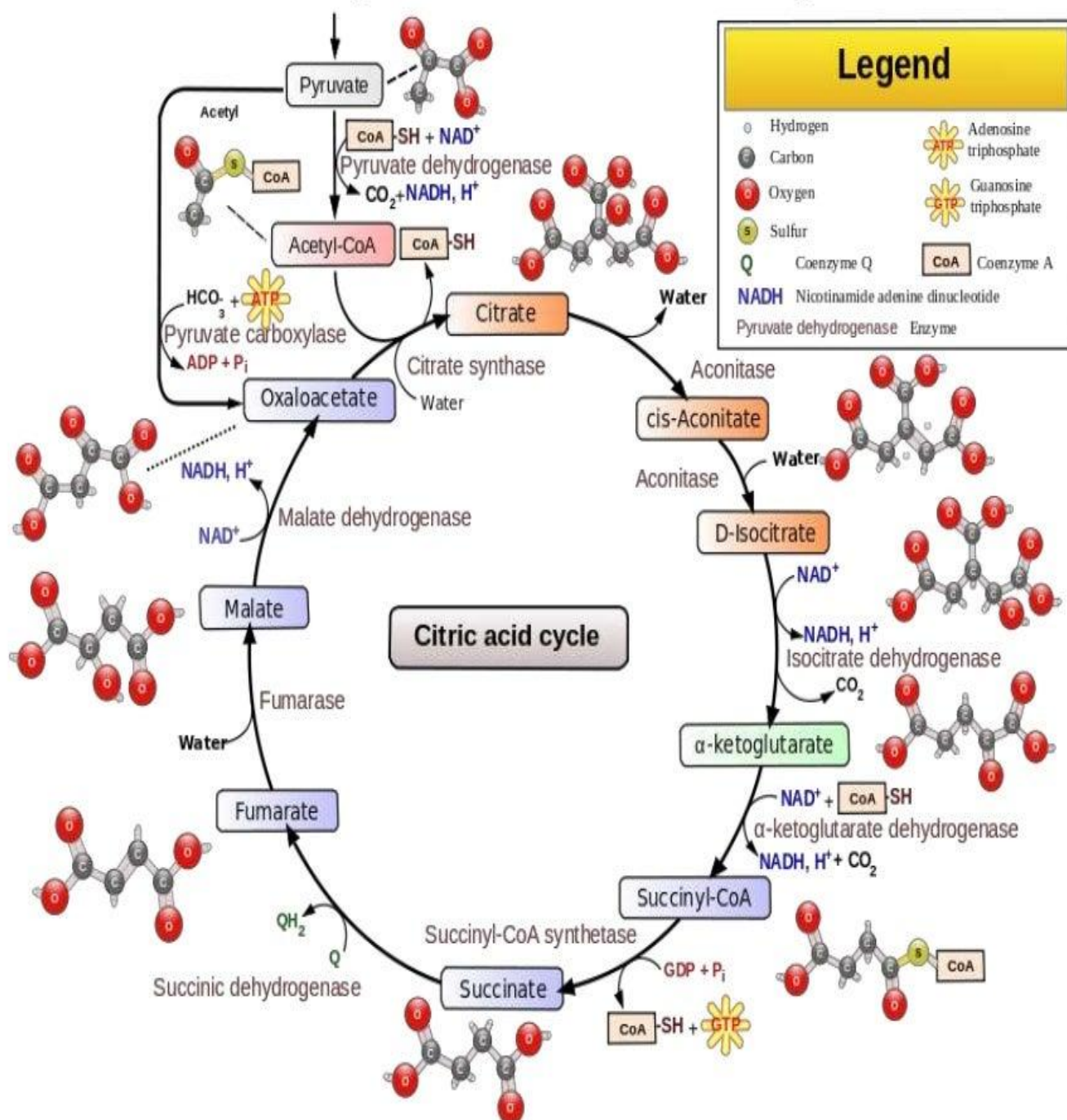
- Glycolysis regenerates **NAD⁺** from NADH during anaerobic conditions through fermentation, ensuring the continuation of glycolytic ATP production.

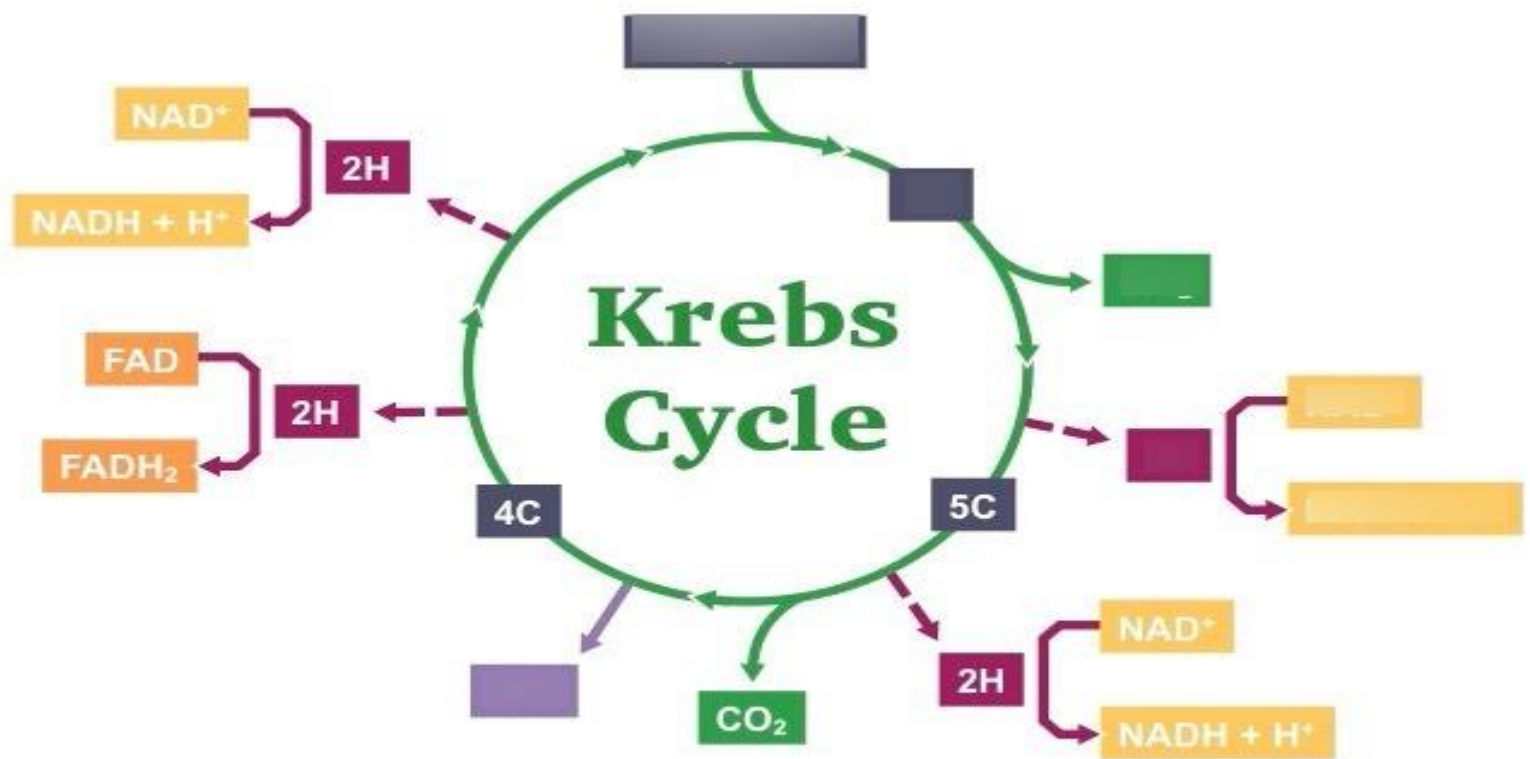
©The Tricarboxylic Acid Cycle

- The tricarboxylic acid (TCA) cycle, also called the **Krebs cycle** or the citric acid cycle, is central to energy metabolism in the body.
- Under aerobic conditions, the TCA cycle is the final pathway by which fuel molecules—carbohydrates, fatty acids, and amino acids—are completely oxidized to CO_2 so that their energy is released and transferred to ATP molecules.
- Greater than 90% of food energy captured and used in the human body involves the TCA cycle in conjunction with oxidative phosphorylation.

- The enzymes of the TCA cycle are located in the mitochondrial matrix and work in concert with “energy carriers” that shuttle high-energy electrons released by the TCA cycle to the electron transport chain located in the inner mitochondrial membrane.
- These socalled energy carriers, NADH and FADH, are formed by reduction (accept electrons) when fuel molecules are oxidized (lose electrons).
- In this way, the main function of the TCA cycle is to generate high-energy electrons that power the synthesis of ATP via oxidative phosphorylation.

TCA Cycle / Krebs Cycle





Single cycle: $2 \times CO_2$; $1 \times ATP$; $1 \times FADH_2$; $3 \times NADH + H^+$

Two cycles: $4 \times CO_2$; $2 \times ATP$; $2 \times FADH_2$; $6 \times NADH + H^+$

@Biological role of TA C

Biosynthetic Precursors

- The cycle provides intermediates for various anabolic pathways:
 - Citrate:** Precursor for fatty acid and cholesterol synthesis.
 - α -Ketoglutarate:** Precursor for glutamate and other amino acids.
 - Succinyl-CoA:** Used in heme biosynthesis.
 - Oxaloacetate:** Precursor for gluconeogenesis and amino acid (e.g., aspartate) synthesis.

Role in Electron Transport Chain (ETC)

- $NADH$ and $FADH_2$ generated by the TCA cycle donate electrons to the ETC, driving ATP synthesis via oxidative phosphorylation.

Regulation

- By producing CO_2 , which is converted to bicarbonate, the TCA cycle contributes to maintaining acid-base balance in the body.

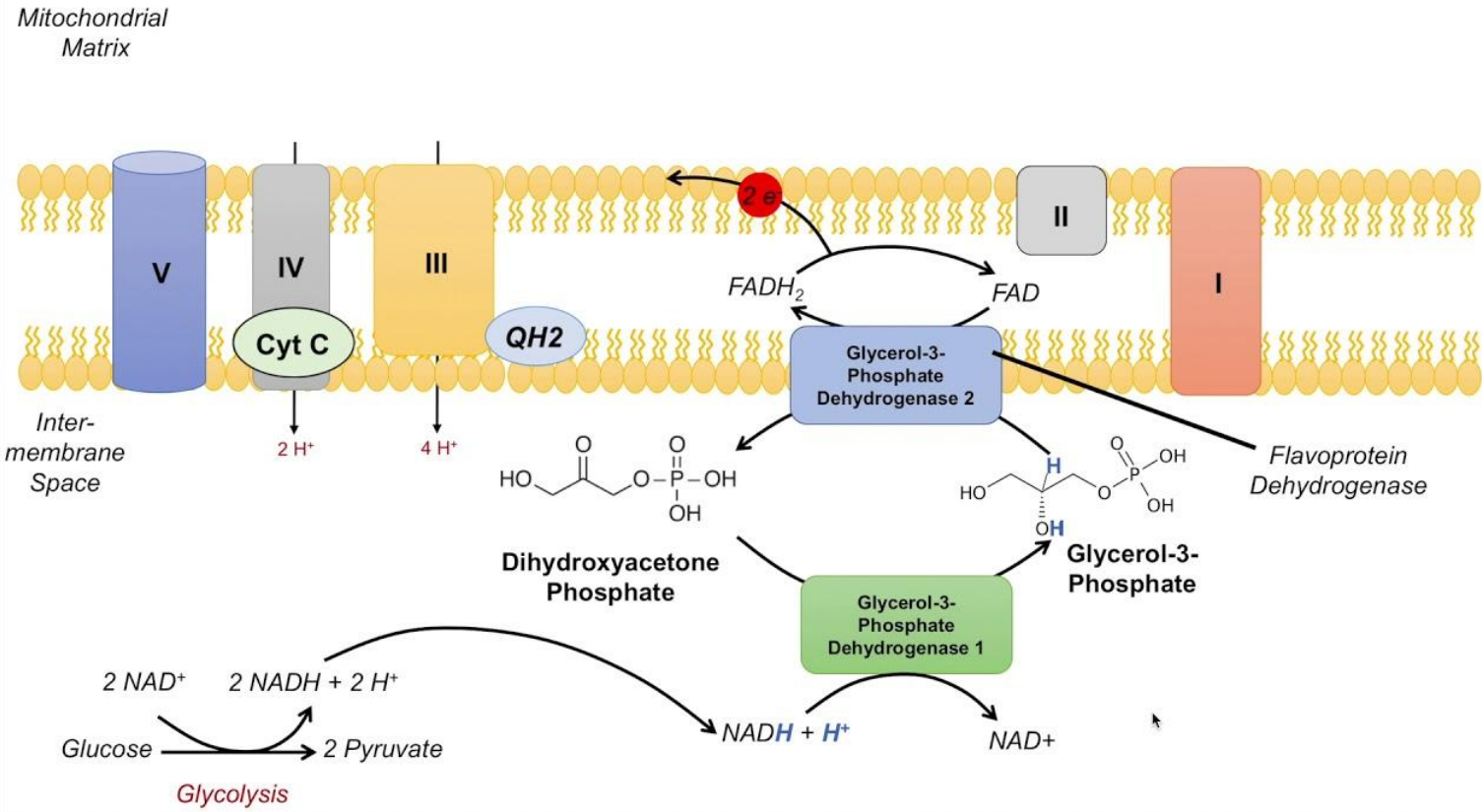
Oxidation of Acetyl-CoA

- The TCA cycle is the final common pathway for the oxidative breakdown of carbohydrates, fats, and proteins.
 - **Glucose** is metabolized into pyruvate (via glycolysis), which is converted into acetyl-CoA.
 - **Fatty acids** undergo β -oxidation to generate acetyl-CoA.
 - **Amino acids** can also feed into the cycle after deamination and conversion to intermediates.

©NADH from Glycolysis: The Shuttle Systems

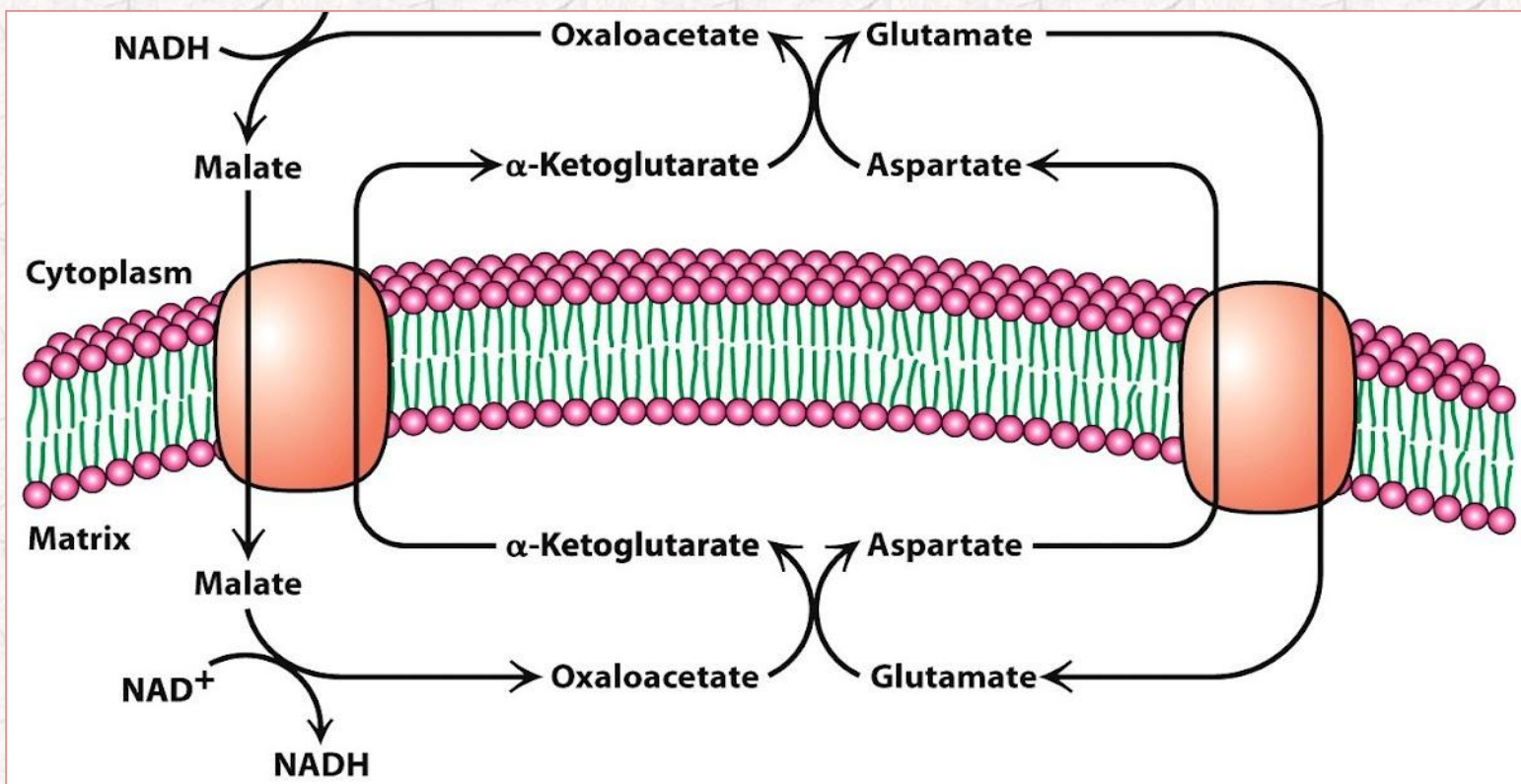
- NADH produced in the cytosol during glycolysis is unable to directly participate in oxidative phosphorylation because the inner mitochondrial membrane is impermeable to NADH.
- Under anaerobic conditions, NADH in the cytosol is used in the lactate dehydrogenase reduction of pyruvate to lactate, thereby becoming reoxidized to NAD^+ without involving oxygen.
- In this manner, NAD^+ is restored to sustain the glyceraldehyde-3-phosphate dehydrogenase reaction, allowing the production of lactate to continue in the absence of oxygen.
- When the supply of oxygen is adequate to allow total oxidation of incoming glucose, the production of pyruvate and NADH from glycolysis is accelerated and lactate is not formed.
- In this situation, the reducing equivalents of NADH (the protons and electrons) are transported from the cytosol to the mitochondrial matrix by two separate shuttle systems.
- These shuttle systems are specific to certain tissues.
- The glycerol-3-phosphate shuttle functions in the brain and skeletal muscle, whereas the more active malate-aspartate shuttle functions in the liver, kidney, and heart.

©Glycerol-3-Phosphate Shuttle System



- NADH in the cytosol transfers its reducing equivalents to dihydroxyacetone phosphate, forming glycerol-3-phosphate that freely diffuses across the outer mitochondrial membrane.
- The reaction is catalyzed by the cytosolic isoform of glycerol- 3-phosphate dehydrogenase. The reducing equivalents of glycerol-3-phosphate are then transferred to FAD that is associated with a membrane-bound isoform of glycerol-3-phosphate dehydrogenase located on the outer face of the inner mitochondrial membrane.
- Finally, the resulting FADH₂ transfers its electrons directly to the electron transport chain, producing 2 ATPs per mole of NADH .
- This shuttle is not reversible

© Malate-Aspartate Shuttle System



- The most active shuttle compound, malate, is freely permeable to the inner mitochondrial membrane. Oxaloacetate from the cytosol is reduced by the NADH to form malate and NAD
- The malate is oxidized by the enzyme malate dehydrogenase to oxaloacetate in the matrix of mitochondria, producing NADH that enters the electron transport chain and generates 3 ATPs per mole of NADH.
- The oxaloacetate undergoes transamination by aspartate amino transferase to form aspartate, which is freely permeable to the inner membrane and can move back out into the cytosol.
- The effect is that reducing equivalents of NADH are transferred into mitochondria, even though the inner mitochondrial membrane is impermeable to NADH itself.
- This shuttle is reversible.

© *Biological Oxidation and the Electron Transport Chain*

- The majority of ATP synthesized in mitochondria begins with the oxidation of fuel molecules and the release of electrons and protons by the TCA cycle.
- The electrons and protons are delivered (by NADH and FADH₂) to the inner mitochondrial membrane where the electrons are passed through a series of intermediate compounds and ultimately to molecular oxygen, which becomes reduced to HO₂ in the process.
- The compounds participating in this sequential reduction-oxidation constitute the electron transport chain, also known as the respiratory chain because the electron transfer is linked to the uptake of O₂, which is made available to the tissues by respiration.
- The energy provided by the electron flow allows the protons to be translocated from the mitochondrial matrix to the space between the inner and outer membranes, which creates an energy gradient that powers the phosphorylation of ADP to form ATP.
- The term *oxidative phosphorylation* is a descriptive blend of simultaneous processes involving electron transport, translocation of protons, oxidation of a metabolite by oxygen, and the phosphorylation of ADP to make ATP.
- Cellular oxidation of a compound can occur by several different reactions: the addition of oxygen, the removal of electrons, and the removal of protons and electrons together (as hydrogen atoms or hydride ions).
- All of these reactions are catalyzed by enzymes collectively termed *oxidoreductases*. Among these, the **dehydrogenases** remove protons and electrons from nutrient metabolites and are particularly important in energy transformation.
- The protons and electrons removed from metabolites by dehydrogenases generally produce NADH or FADH₂, which are either already in or shuttled into the mitochondria and move along the electron transport chain.
- After oxidation of substrate molecules by a dehydrogenase enzyme, the protons and electrons are transferred to a cosubstrate, such as the vitamin-derived nicotinamide adenine dinucleotide 1 (NAD) or flavin adenine dinucleotide (FAD).
- After accepting protons and electrons from reactions of the TCA cycle, NADH and FADH₂ move to the inner mitochondrial membrane to initiate the electron transport chain.

©Components of the Electron Transport Chain and Oxidative Phosphorylation

Complex I:

- **NADH-Coenzyme Q Oxidoreductase** Complex I— also known as NADH dehydrogenase—transfers a pair of electrons from NADH to CoQ. The structures of the oxidized and reduced forms of CoQ .
- Complex I is made of many polypeptide chains, a molecule of FMN, and several Fe-S clusters, along with additional iron molecules.
- The iron molecules bind with the sulfurcontaining amino acid cysteine.
- The iron transfers one electron at a time, cycling between Fe²⁺ and Fe³⁺.
- CoQ is a highly hydrophobic compound and it diffuses freely in the hydrophobic core of the inner membrane.
- The result of the multistep reaction is the transfer of electrons and protons from NADH to CoQ to form first CoQ hydroquinone and then CoQH₂ and actively transfer protons from the matrix side of the inner mitochondrial membrane to the intermembrane space.
- The importance of the buildup of protons in the intermembrane space is discussed in the following sections.
- The oxidation of NADH through the electron transport chain results in the synthesis of approximately 3 ATP molecules.

Complex II:

- **Succinate Dehydrogenase** Complex II is the succinate dehydrogenase enzyme, which is the only TCA cycle enzyme that is an integral part of the inner mitochondrial membrane.
- Beside the succinate dehydrogenase, Complex II contains a FAD protein and Fe-S clusters (similar to those discussed previously).
- When succinate is converted to fumarate in the TCA cycle, FAD is reduced to FADH₂. The FADH₂ is oxidized with one electron transfer through the Fe-S centers to reduce CoQ to CoQH₂.
- Unlike in Complex I, the protons released from FADH₂ are not transported to the intermembrane space.
- The oxidation of FADH₂ through the electron transport chain results in the formation of approximately 2 molecules of ATP.

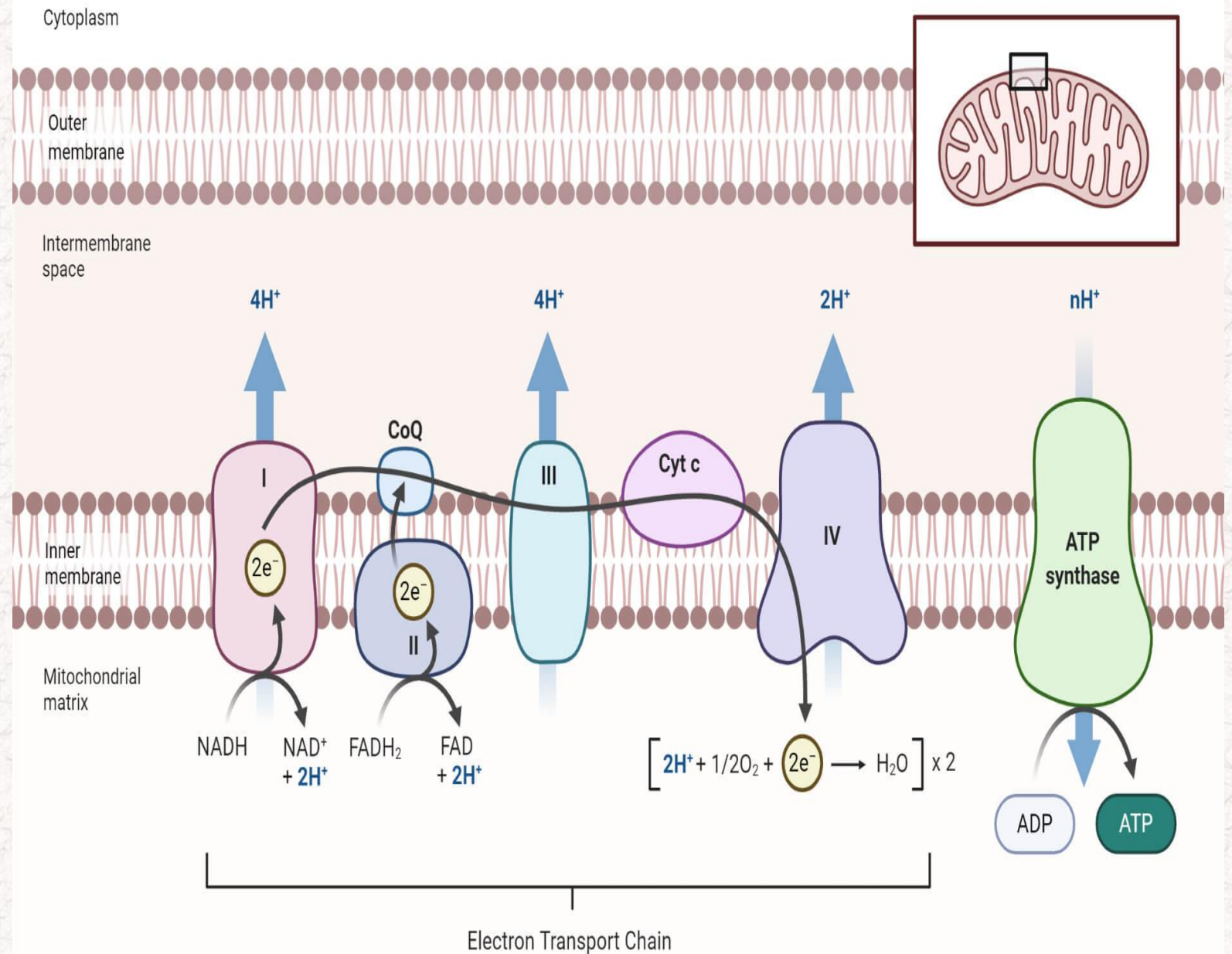
Complex III:

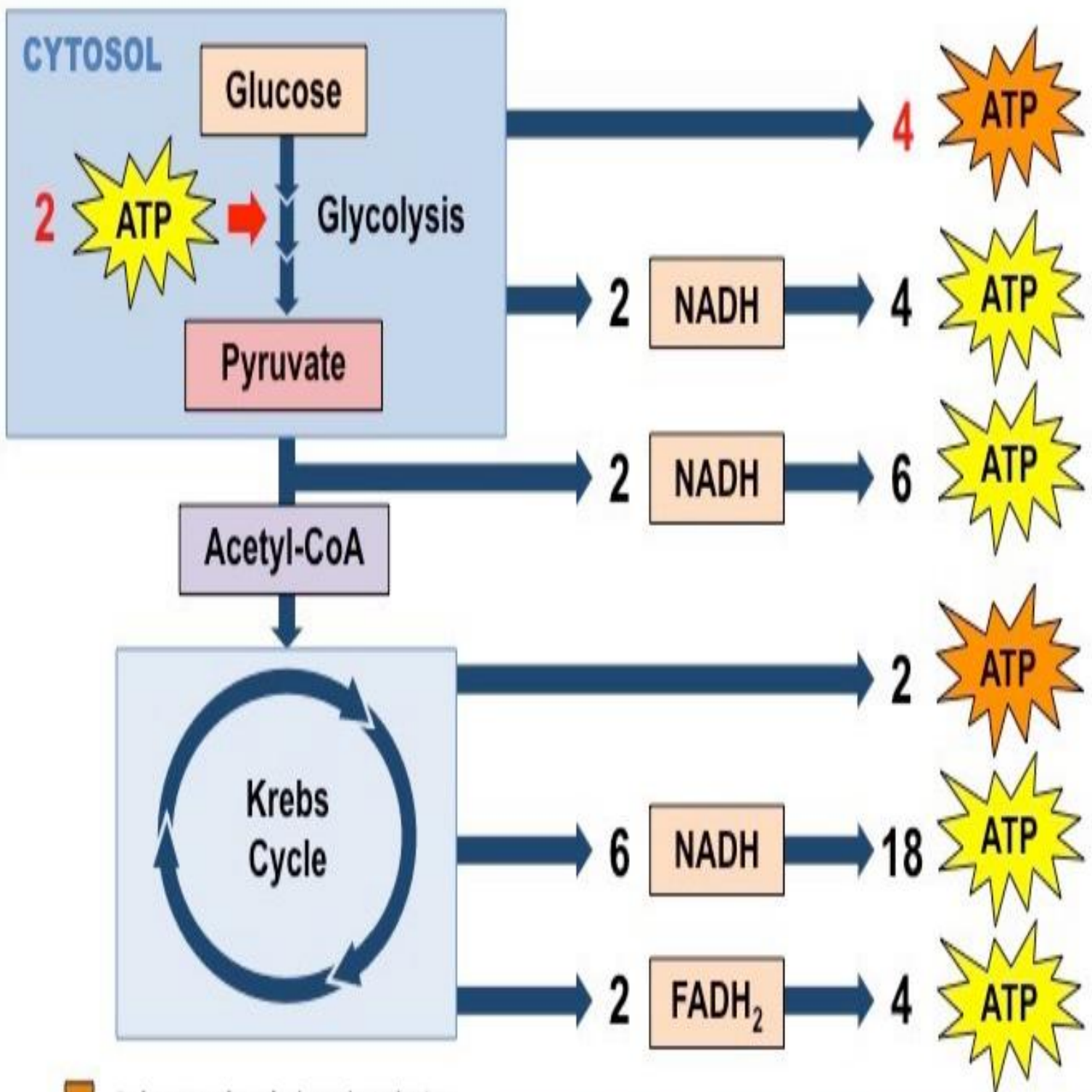
- **Coenzyme Q-Cytochrome c Oxidoreductase** Reduced CoQ passes its electrons to cytochrome c in the third complex of the electron transport chain in a pathway known as the Q cycle.
- The complex contains three different cytochromes and Fe-S protein.
- The cytochromes contain heme molecules with an iron molecule in the center.
- The iron in the center of the cytochromes is oxidized and reduced as electrons flow through.
- Electrons pass through the Q cycle in two phases; In the first phase a CoQH₂ (**ubiquinol**) passes one electron to form the semiquinone (one of the two hydroquinones oxidized), then another electron and proton are transferred to the semiquinone to produce oxidized CoQ (quinone), releasing four protons to the intermembrane space
- The electrons are then transferred to cytochrome c₁ (and associated cytochromes), and CoQ picks up two protons from the matrix, resulting in the reduction of a CoQ to CoQH₂.
- This means that two turns of the CoQ cycle result in the oxidation of 2CoQH₂ to CoQ, the release of 4H⁺ in the intermembrane space, and the reduction of one CoQ to CoQH₂.
- Like CoQ, cytochrome c is a mobile carrier.
- This characteristic means that cytochrome c is able to migrate along the membrane.
- Cytochrome c associates loosely with the inner mitochondrial membrane on the matrix side of the membrane
- It can then pass its electrons on to cytochrome c oxidase in Complex IV.

Complex IV:

- **Cytochrome c Oxidase** Complex IV is called cytochrome c oxidase.
- It accepts electrons from cytochrome c and catalyzes a four-electron reduction of oxygen to form water.
- This reaction is the final one in the oxidation of the energy-providing nutrients (carbohydrate, fat, protein, and alcohol) to produce usable chemical energy in the form of ATP.
- The structure of cytochrome c oxidase is known; it is made up of multiple subunits.
- Some of the subunits are encoded from nuclear DNA and some from mitochondrial DNA.

- These latter proteins contain iron and copper.
- These metal ions cycle between their oxidized (Fe^{3+} , Cu^{2+}) and reduced (Fe^{2+} , Cu^{1+}) states.
- Cytochrome c oxidase also contains two cytochromes, cytochrome a and cytochrome a₃, which contain different heme moieties.
- Protons are transported to the intermembrane space.
- Electron transport can carry on without phosphorylation, but the phosphorylation of ADP to form ATP is dependent upon electron transport, with the transport terminating as molecular oxygen is reduced to H_2O .





 *Substrate level phosphorylation*

 *Oxidative phosphorylation*

Total net ATP yield: 36

